

CIS 635: Assignment 1

Exploratory Data Analysis of Climate Change Data

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Objective and Logistics

Objective: hands-on practice with

- Data preparation
- Descriptive statistics
- Exploratory data analysis (EDA)
- Data visualization

Work with climate-change data; use numerical summaries and visualizations together to identify patterns, variability, and relationships.

At a glance

Total points: 10

Deadline: check the Blackboard post

The data are samples — do not make global conclusions from them.

Use the following two files:

Climate distortion data

```
.../data_mining_kdd/refs/heads/main/data/climate-change/  
climate_distortion.csv
```

Country-region data

```
.../data_mining_kdd/refs/heads/main/data/climate-change/  
country_region.csv
```

Both files are hosted on the course GitHub ([mdkamrulhasan/data_mining_kdd](https://github.com/mdkamrulhasan/data_mining_kdd)).

(a) Data Preparation and Initial EDA [1.5 pts]

- (i) Identify the number of columns containing NULL values in either data file and briefly describe the pattern of missingness
- (ii) Impute the NULL values using a reasonable method/logic and briefly justify your choice
- (iii) Report the number of unique Region(s) recorded for the USA

(b) Descriptive Statistics [2.5 pts]

For the USA data, analyze CO2 Emissions MT and Pesticide Use KG per HA over the available period.

- (i) Report the mean, median, variance, standard deviation, minimum, and maximum
- (ii) Report Q_1 , Q_3 , and the interquartile range: $IQR = Q_3 - Q_1$
- (iii) Briefly compare the two variables using the numerical summaries — what do the statistics suggest about their center and variability?

(c) Exploratory Data Analysis and Relationships [3 pts]

- **(i) 1** Which country had the most and the least CO2 Emissions MT before and after 2000? Report the values
- **(ii) 0.5** What was the coolest date recorded in the USA?
- **(iii) 1** Which Crop Type from the “Midwest(U)” and “Northeast(U)” regions together had the highest historical Economic Impact Million USD?
- **(iv) 0.5** For the eight variables below, identify the pair with the highest association and the pair with the lowest. Explain what each association suggests

Average Temperature C, Total Precipitation mm, CO2 Emissions MT, Extreme Weather Events, Pesticide Use KG per HA, Fertilizer Use KG per HA, Soil Health Index, Economic Impact Million USD

(d) Visualization and Interpretation [3 pts]

Create clear, properly labeled visualizations that support your EDA.

- **(i) 1** Histogram for each of CO2 Emissions MT and Pesticide Use KG per HA for the USA — describe the distribution shape (symmetric, skewed, bimodal, heavy-tailed, ...)
- **(ii) 1** A graph of the yearly average Total Precipitation mm in the USA — describe any visible trend or pattern
- **(iii) 1** A scatter plot for one variable pair from part (c)(iv) — use it to support your interpretation of the association

Points Breakdown

Task	Points
(a) Data Preparation and Initial EDA	1.5
(b) Descriptive Statistics	2.5
(c) Exploratory Data Analysis and Relationships	3.0
(d) Visualization and Interpretation	3.0
Total	10.0

Submission Guidelines

- Submit a **runnable notebook** (Python, R, or another language of your choice) through Blackboard
- A link to a notebook kept in Google Drive or elsewhere is **not accepted**
- Include sufficient markdown/comments to explain your analysis and decisions
- Use appropriate labels, titles, units, and legends in visualizations
- Report numerical results to **three decimal places** where appropriate

Important

The goal is **not** full test-style coverage of every possible data operation. Focus on meaningful analysis: use descriptive statistics, EDA, and visualization **together** to support your observations and conclusions.